

IS 477 Project Interim Progress Report - Siddhu Sami and Colin Papp

Project Overview: [IS477-FA25-Course-Project-Instructions - Google Docs](#)

Individual Member Contribution Reports Submitted by Both Members Separately

Project Pivot:

- Our initial plan was to analyze spending allocation by position for NFL teams and seeing how it correlates with various team success metrics, essentially finding the what spending allocations to certain positions is related to the most team success.
- Due to a dearth of data that was freely available, complete, and reputable for NFL player salary data, we pivoted to doing the same exact thing but for NBA teams
- This resulted in much more reliable, accessible, complete, and clean data that was easier to process and analyze without significant web scraping

Member Contributions

- Colin: found both datasets, worked on cleaning the team results datasets and creating preliminary visualizations
- Siddhu: cleaned salary data, calculated team spending allocation and linked it to team result dataset

Salary Data: from information science professional Edwin Jeon, who cleaned data from data scientist Erik Gregory Webb. Both individuals are full time data science professionals who have done this recreationally, and share their datasets for educational purposes, allowing free use, as stated in their Github pages. This dataset relevant columns we will use here are Player, Year, Salary, Position and Team. This is the dataset we will use to derive spending allocation for each team.

- Kaggle Link:
<https://www.kaggle.com/datasets/ratin21/nba-player-stats-and-salaries-2000-2025?resource=download>
- Original Data: <https://github.com/erikgregorywebb/datasets/blob/master/nba-salaries.csv>

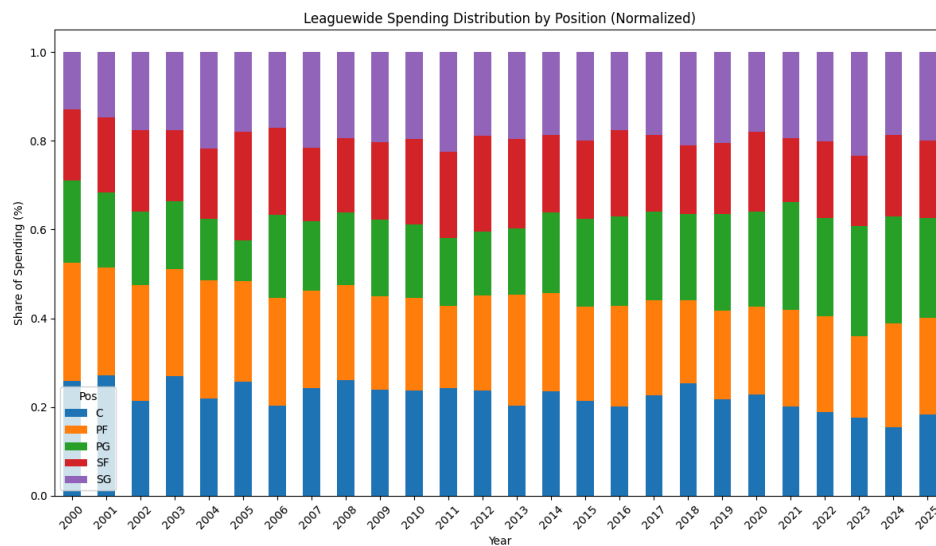
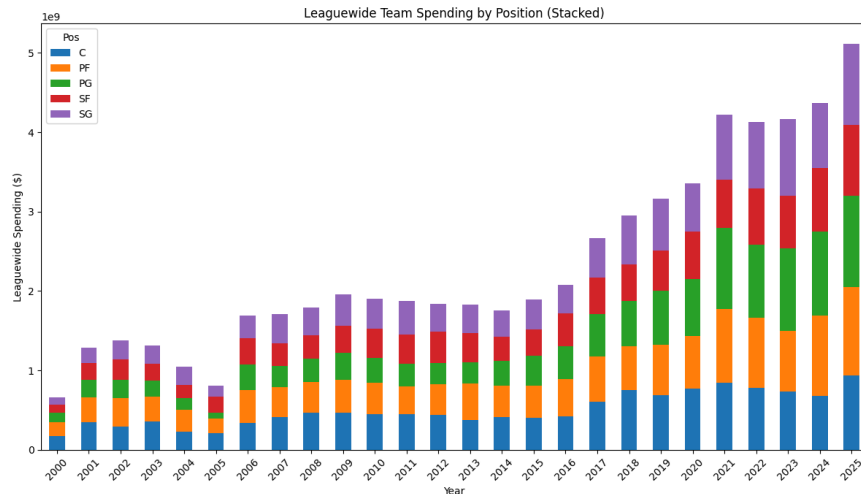
Cleaned
Data: https://github.com/edwinjeon/NBA-Salary-Prediction/blob/main/data/NBA%20Player%20Stats%20and%20Salaries_2000-2025.csv
- Cleaning Methodology:
<https://github.com/edwinjeon/NBA-Salary-Prediction/blob/main/NBA%20Salary%20Project%20Data%20Preparation.ipynb>
- Statement of Free Use: <https://github.com/erikgregorywebb/data?tab=MIT-1-ov-fil>

Salary Data Processing:

- The salary data was filtered to only contain the relevant columns of "Player", "Salary", "Pos", "Year", "Team"
- A small but significant portion of players were listed as dual position (e.g. SF-SG instead of SF). We made the decision to assign these players to the first position listed
- Then, salary counts were grouped by position and team and normalized, resulting in each team, each season, having a salary distribution by position to capture spending allocation
- This spending allocation data was then linked to the season/team results dataset, so we can eventually compare spending allocation to team success

Salary Data Preliminary Exploratory Data Analysis

- To explore trends in data before more computational work, we created visualizations to seek out trends in the data
- The first chart is a stacked bar chart of league-wide spending by position, by year
- The second chart is the first chart normalized since league spending on players has dramatically increased over time
- Both charts go back to 2000, but only 2005 and on will be used for further analysis
- We can see that a lower portion of player spending has gone towards centers and power forwards, with an increasing share going to guards. This can likely be attributed to the increasing importance of three point shooting in the league.



Season Data: from [craftedNBA.com](https://craftednba.com), a website for curated data for professional basketball fans. For our project, we used their NBA season data that spans back to 1955 and contains dozens of stats for each team, each season with stats including but not limited to Team Strength, Wins, Losses, Age (average age of players), SRS (strength of schedule impact), offensive/defensive ratings, true shooting, pace, and playoff wins and losses. Because we don't have a VIP membership to the website, we had to manually copy and paste data into the excel. We did not edit the excel, only used it as a medium to access the data after it was pasted in and converted to CSV for further analysis. This is the dataset we will use to compare against spending allocation.

- [Season Data.xlsx](#)
- [Historical NBA Team Stats | CraftedNBA | NBA Stats & Analytics](#)

- The dataset is comprised of factual, non-copyrightable NBA statistical information (wins, losses, efficiency metrics) and is being used solely for non-commercial analytical and educational purposes

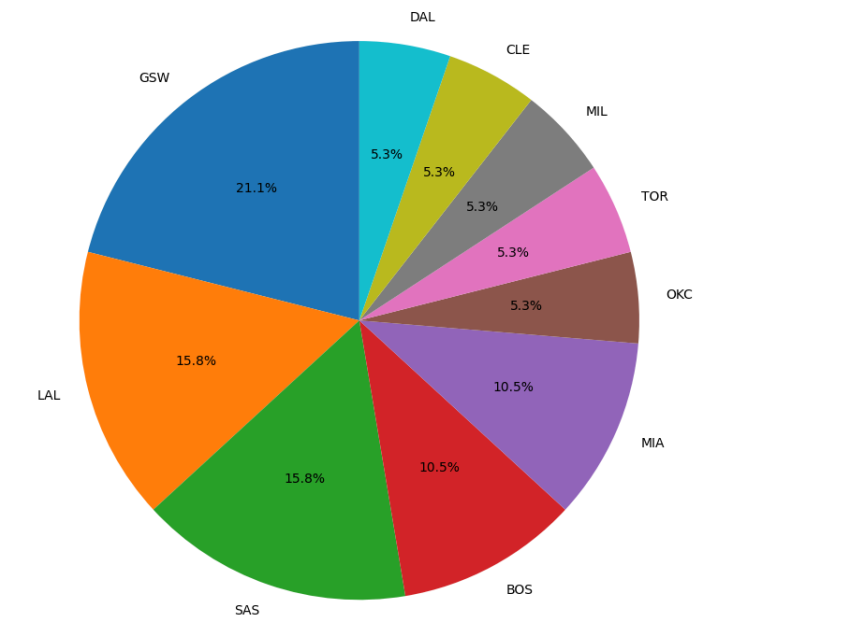
Season Data Processing:

- The data was filtered to exclude most of the playoff advanced stats, only including playoff wins and losses, it was also filtered to only include 2005 and beyond.
- Blank playoff wins and losses columns were filled with zeros, as these columns were intended to be left blank since the teams missed the playoffs.
- A boolean 0/1 variable was created for both playoff appearance and championship win, if a season had playoff wins/losses listed it was marked as a playoff appearance. From the initial data, it had an emoji by the team's name if they won that season which was read as question marks, so if a team had question marks in there name they were marked as winning the championship and the question marks were removed.
- Since teams enter and exit the NBA, the data was filtered for teams that have at least 17 seasons represented in the dataset
- The data was checked for na values, and none remained
- The salary dataset used abbreviated team names while the season data used full names (e.g. CLE instead vs. Cleveland Cavaliers). To fix this, a name map was created matching the full names to the abbreviated versions, and team names in the season dataset were mapped to match the names in the salaries dataset

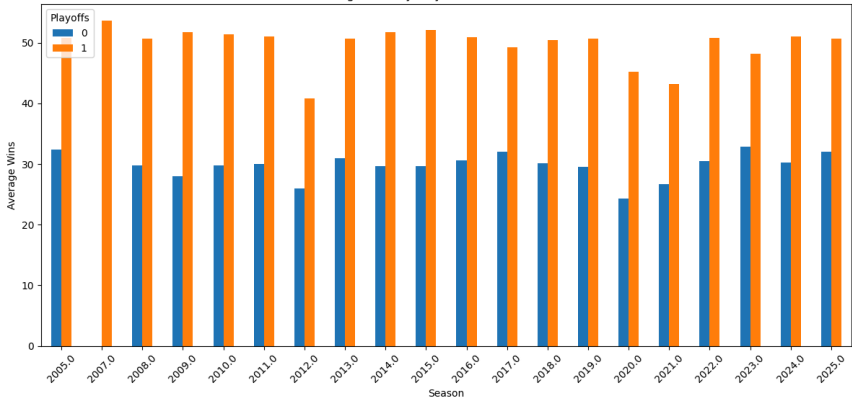
Season Dataset Preliminary Exploratory Data Analysis

- For the purpose of exploratory data analysis, we created the following graphs using Matplotlib to see the distribution of championships, average wins for playoff vs. non playoff teams by year, and average win percentages for teams across the dataset. This was done using `value_counts()`, `.groupby()`, and creating a win percentage column

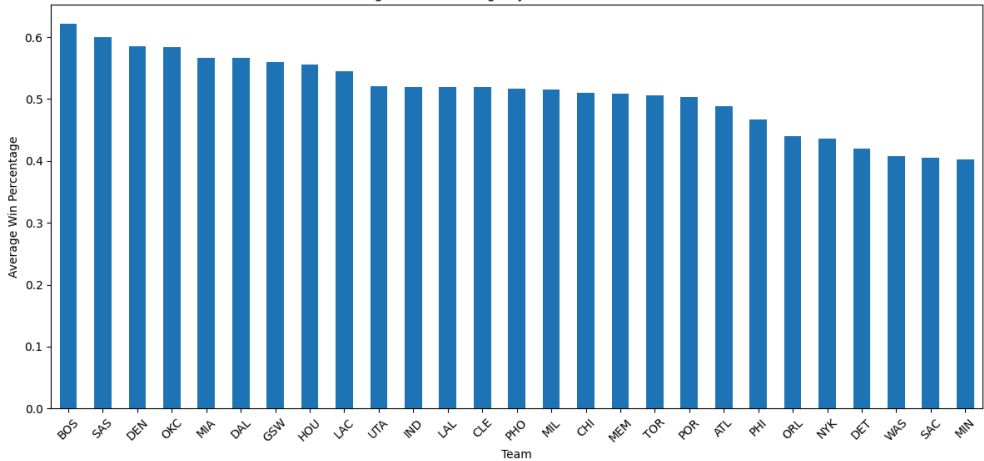
Distribution of Championship Winning Teams



Average Wins by Playoff Status Each Season



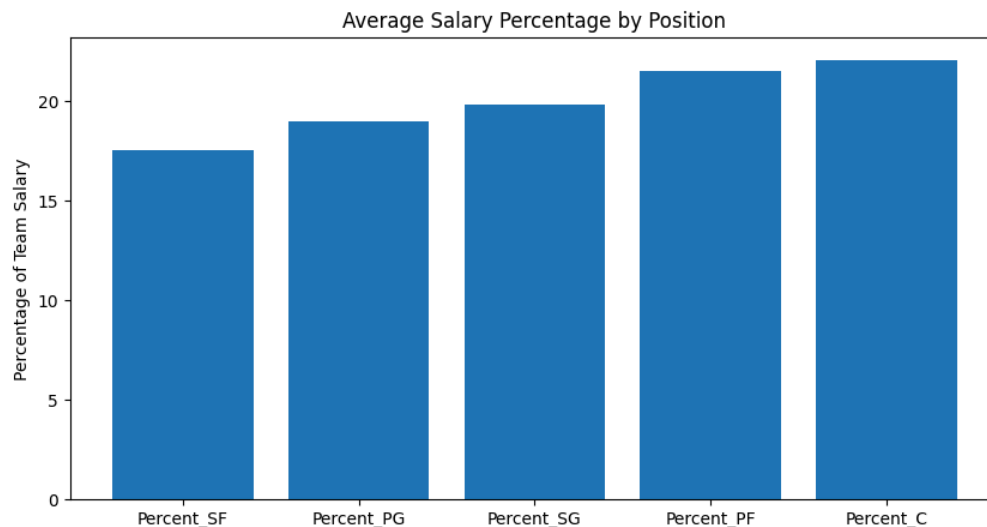
Average Win Percentage by Team Across All Seasons

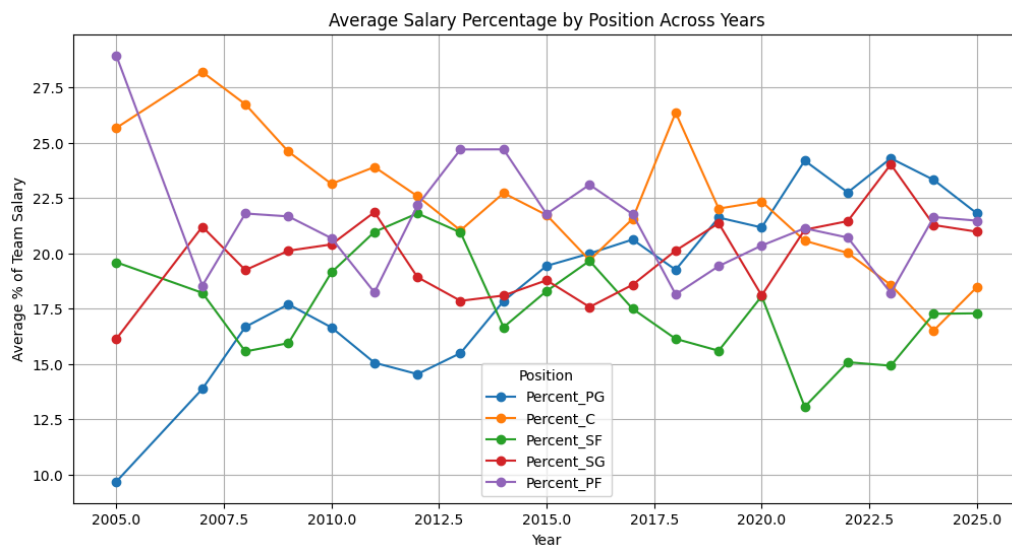
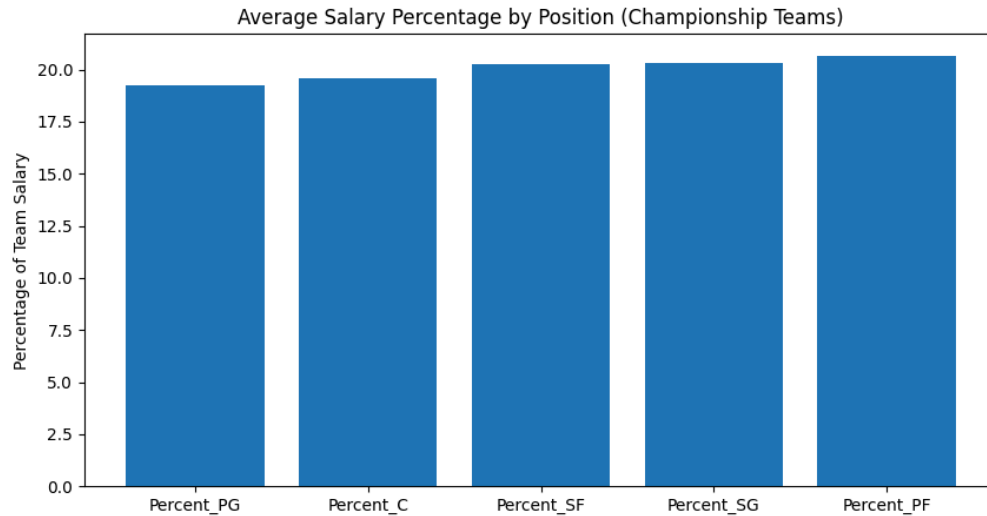


Merged Data: We created a merged pandas dataframe of the salary data and the seasons data. This data frame includes every season from every NBA team that has 17 seasons in the dataset from 2005 onwards and includes the percentage of their total salary spent on each position. This dataframe will be used to perform analysis on NBA teams' salary allocation and seek to find relations between this and team success. Which can be determined by multiple different metrics including regular season wins, regular season offensive and defensive performance, overall player performance, and if the team won an NBA Championship.

Merged Dataframe Exploratory Data Analysis

- We made sure that there were no null values that may have popped up during the process of merging the two datasets.
- We noticed that there were some rows where one or more of the columns with percentage spent were 0. We decided not to remove these columns because this indicate a team decision to not spend on that position
- We also created visualizations using Matplotlib. The first one was created in order to see the average percentage of team salary allocated to each position over all seasons and every team. We also did the same thing for only teams that won the championship that season. Lastly we created a graph that shows the average all NBA teams' percentage of a team's salary that was allocated to each position over every season that was included in our dataset which was from 2005-2025





What's Next:

- Next, we will seek to perform multiple linear regression analysis using sci-kit learn, LASSO and Ridge regression techniques to find correlations between the positional player spending allocation and various season by season team success metrics
- We will analyze the strength and nature of these relationships and continue to analyze assumptions and feature selection
- Our end goal is to determine what optimal spending mixes have been throughout the past 20 NBA seasons, how that mix has changed over time, and attempt to extract insights on how NBA teams can spend to maximize their competitiveness